

CATALYSIS

Disentangling the activity-selectivity trade-off in catalytic conversion of syngas to light olefins

Feng Jiao^{1,2†}, Bing Bai^{1,2†}, Gen Li^{1,2†}, Xiulian Pan^{1,2*}, Yihan Ye^{1,2}, Shengcheng Qu¹, Changqi Xu¹, Jianping Xiao^{1,2}, Zhenghao Jia¹, Wei Liu¹, Tao Peng^{1,3}, Yilun Ding^{1,2}, Cheng Liu^{1,2}, Jinjing Li¹, Xinhe Bao^{1,2,3*}

Breaking the trade-off between activity and selectivity has been a long-standing challenge in the field of catalysis. We demonstrate the importance of disentangling the target reaction from the secondary reactions for the case of direct syngas conversion to light olefins by incorporating germanium-substituted AlPO-18 within the framework of the metal oxide–zeolite (OXZEO) catalyst concept. The attenuated strength of the catalytically active Brønsted acid sites allows enhancing the targeted carbon-carbon coupling of ketene intermediates to form olefins by increasing the active site density while inhibiting secondary reactions that consume the olefins. Thus, a light-olefins selectivity of 83% among hydrocarbons and carbon monoxide conversion of 85% were obtained simultaneously, leading to an unprecedented light-olefins yield of 48% versus current reported light-olefins yields of $\leq 27\%$.

Achieving both high activity and high selectivity is a long-standing challenge for selective reactions such as the hydrogenation of CO in direct syngas conversion (1–8). Fischer-Tropsch synthesis (FTS), as a key process for syngas conversion using Fe-based (8–10) and Co-based (11–13) catalysts, allows direct synthesis of light olefins (two to four carbon atoms, $C_2=C_4$), named as Fischer-Tropsch-to-olefins (FTTO). Substantial progress has been made in the past decade (table S1). For example, a 61% light-olefins selectivity among hydrocarbons at 0.5 to 1% CO conversion and 52% selectivity at 88% CO conversion had been obtained, giving 26% yield of light olefins over Fe/carbon nanofiber catalysts (8). Hydrophobically modified FeMn catalysts gave 63% selectivity for all olefins and $\sim 27\%$ for light olefins at 56% CO conversion, amounting to $\sim 14\%$ yield of light olefins (9). A light-olefins selectivity of 61% among hydrocarbons

was achieved at 32% CO conversion over Mn-promoted Co_2C catalysts, giving 10% $C_2=C_4$ yield (11). More recently, CO conversion was enhanced to 64%, with 71% light-olefins selectivity among hydrocarbons when CoMn carbide was physically mixed with hydrophobic polydivinylbenzene, giving 24% yield of light olefins (13).

In addition, bifunctional metal oxide–zeolite (OXZEO) catalysts were recently developed for direct syngas conversion to light olefins (6, 7, 14–16). By separating the key steps of CO and H_2 activation from the subsequent C–C coupling of the ketene intermediates (CH_2CO , which is a carrier of CH_2) on the active sites of metal oxides and zeolites (or zeotypes), respectively, the product selectivity can be modulated through the topology of zeolites (or zeotypes), and light-olefins selectivity higher than 80% was obtained (6, 7, 16). Recently, the technology was successfully demonstrated in a pilot plant with 1000 ton/year light-olefins

output in Shaanxi, China (7, 17). Extensive efforts have been made to optimize the catalysts by screening a wide range of metal oxide and zeolite (zeotype) combinations as well as reaction conditions (7, 16, 18–24). As a result, the catalytic performance has been greatly improved; CO conversion could be increased from 17 to 60% with light-olefins selectivity being maintained at 75%, and the yield of light olefins improved from 8 to 27% (16, 21, 24). However, further improvement of OXZEO and FTTO processes appears to be stalled by an activity-selectivity trade-off (Fig. 1A, dashed line, and table S1).

The mechanistic origin of the activity-selectivity trade-off can be traced to the catalytic sites for both the target and side reactions on the catalyst. Taking the silicoaluminophosphate (SAPO)-type zeotypes in the OXZEO syngas conversion concept as an example, the Si–OH–Al Brønsted acid sites are the active sites, catalyzing C–C coupling of the ketene intermediates to form light olefins as products (7, 18), which consequently shifts the thermodynamically limited step of syngas conversion to ketene (7, 25). A high selectivity of light olefins has been achieved by weakening the acidity through reduction of the Si content (7, 16, 18). However, this reduces simultaneously the density of the Si–OH–Al acid sites and hence the rate of intermediate consumption and ultimately CO conversion. CO conversion can be facilitated with a higher acid site density by increasing the Si content of the SAPO zeotypes. But this comes at the expense of olefins selectivity because the Si–OH–Al Brønsted acid sites are

¹State Key Laboratory of Catalysis, 2011-Collaborative Innovation Center of Chemistry for Energy Materials, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China. ²University of Chinese Academy of Sciences, Beijing 100049, China. ³University of Science and Technology of China, Hefei 230026, China.

*Corresponding author. Email: panxl@dicp.ac.cn (X.P.); xhbao@dicp.ac.cn (X.B.)

†These authors contributed equally to this work.

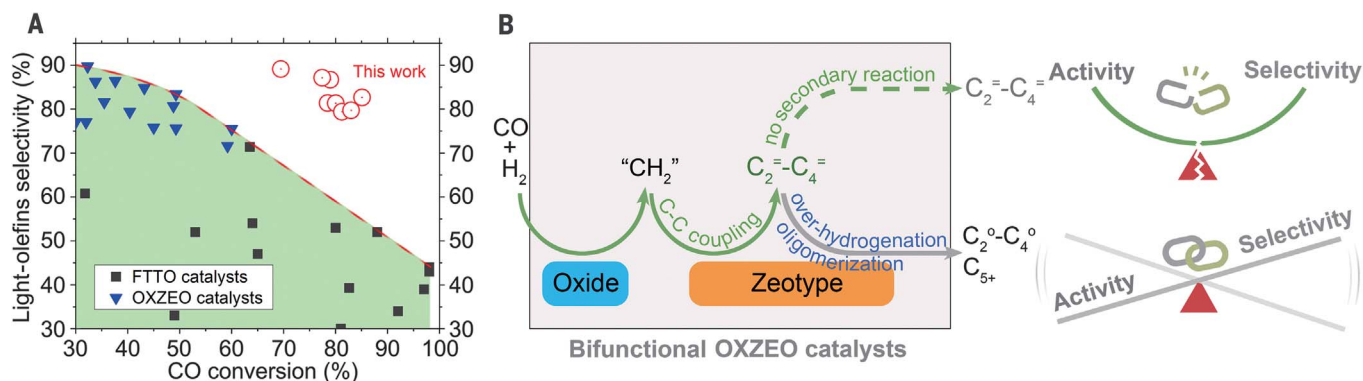


Fig. 1. The activity-selectivity trade-off in the OXZEO syngas-to-light olefins process. (A) Selectivity of light olefins versus CO conversion data reported for FTTO (black squares) and OXZEO (blue triangles) processes (6–14, 16, 18–24, 30), and this work (red circles). The red dashed line is a guide for the eye. (B) Scheme depicting the activity-selectivity trade-off caused by the entangled target (ketene intermediate-to-light olefins) and undesired secondary reactions (overhydrogenation to paraffins and oligomerization to C_{5+} hydrocarbons) over zeotypes.

also active for the secondary reactions of olefins, such as hydrogenation and oligomerization (18, 26–28). Moreover, these secondary reactions are enhanced by the simultaneously strengthened acidity and increased active sites (6, 7, 14, 16, 18, 29). For example, Li *et al.* showed that CO conversion was enhanced from 27 to 50% by increasing the Si content of SAPO-18, but the light-olefins selectivity declined from more than 80% to around 50% because of the simultaneous acceleration of secondary hydrogenation reactions (18). It remains a mechanistic challenge for conventional aluminosilicate and SAPO zeotypes.

Therefore, the goal is to terminate the reaction at the olefin formation step—that is, to disentangle the target reaction C–C coupling of ketene from the secondary reactions of hydrogenation and oligomerization of olefins (Fig. 1B). We report that Ge-substituted AlPO-18 is an effective strategy to achieve this goal and thus break the activity-selectivity trade-off. As a result, we obtained an unprecedented CO conversion of 85% simultaneously with 83% light-olefins selectivity among hydrocarbons. The light-olefins yield reaches as high as 48%, which nearly doubles the best values reported previously for FTTO and OXZEO processes.

Germanium incorporation in AlPO-18

We introduced Ge into AlPO-18 by means of a hydrothermal method, with the Ge/Al ratio varying between 0.012 and 0.027 (denoted as the subscript “a”). The resulting GeAPO-18_a zeotypes exhibited a typical AEI crystal structure with no impurity phase and homogenous distribution of Ge species throughout the crystals (figs. S1 to S4 and table S2). Taking GeAPO-18_{0.025}, for example, x-ray absorption spectroscopy (XAS) of the Ge K-edge (fig. S5 and table S3) indicated that Ge species were dispersed as single sites being fourfold coordinated with O atoms within the AlPO-18 framework. Infrared analysis of the hydroxyl (OH) stretching vibration (fig. S6) and NH₃ adsorbed on GeAPO-18_a (fig. S7) confirmed the generation of Ge–OH–Al Brønsted acid sites created by the framework substitution of P by Ge atoms (figs. S8 and S9). A hydrothermal treatment experiment at 450°C demonstrated a good stability (fig. S10). NH₃ temperature-programmed desorption (TPD) indicated that the acid site density increased with the Ge/Al ratio up to 0.027, but the acid strength remained about the same (fig. S11 and table S4).

The GeAPO-18_a variants were physically mixed with ZnCrO_x to form OXZEO composite catalysts for syngas conversion. CO conversion reached 60% upon introduction of a small amount of Ge (Ge/Al = 0.012), and light-olefins selectivity among hydrocarbons was 84% at 410°C. CO conversion rose almost linearly to 79%, with the density of the medium-strength acid sites increasing to 53 μmol/g,

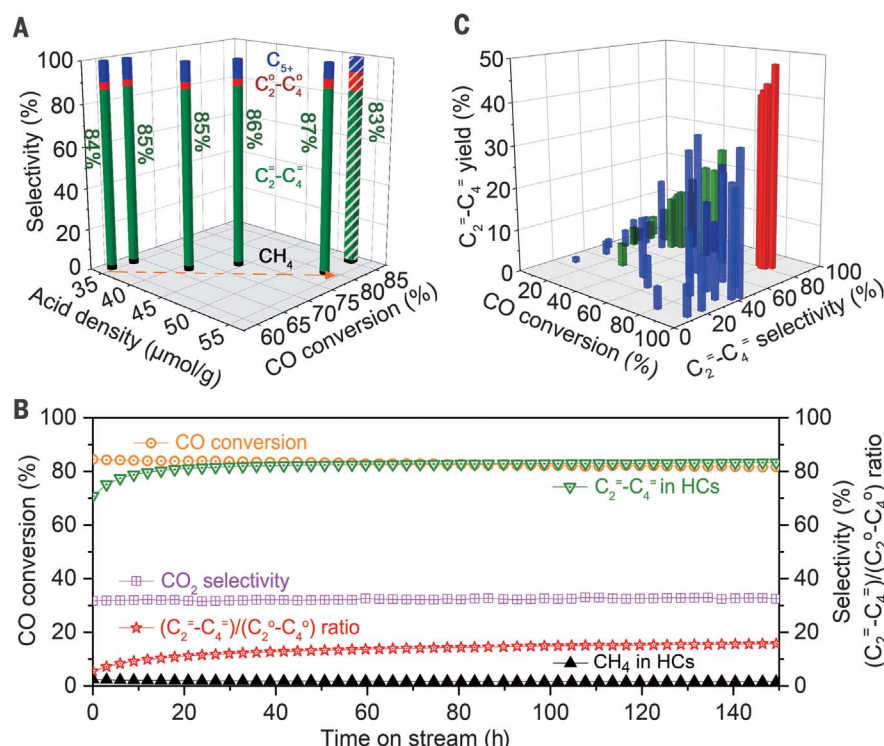


Fig. 2. Catalytic performance of ZnCrO_x-GeAPO-18 catalysts in syngas conversion. (A) Performance as a function of the medium-strength Brønsted acid site density of GeAPO-18. The dashed arrow line is a guide for the eye. Reaction condition: ZnCrO_x/GeAPO-18 = 2 (mass ratio), H₂/CO = 2.5, 410° and 430°C (prism filled pattern), 6 MPa and 1500 ml/(g_{cat}·hour). (B) Stability test of ZnCrO_x-GeAPO-18_{0.027} at 430°C. (C) Performance of ZnCrO_x-GeAPO-18 (red columns) in comparison with previously reported FTTO (blue) and OXZEO catalysts (green) (6–14, 16, 18–24, 30).

and light-olefins selectivity increased to 87% (Fig. 2A). CO conversion further increased to 85% at 430°C, and light-olefins selectivity declined only slightly to 83% (Fig. 2A and fig. S12). By contrast, AlPO-18 containing no Ge exhibited a low CO conversion of 24% and light-olefins selectivity of 63% under the same conditions (fig. S13 and table S5).

A test of 150 hours on stream showed that the catalyst delivered stable performance during this time frame (Fig. 2B, fig. S14, and table S6). The ratio of light-olefins to light-paraffins, denoted as (C₂=C₄)/(C₂^o-C₄^o), remained at 16. The yield of light olefins reached as high as 48% (Fig. 2C), surpassing the previously reported best values for both FTTO and OXZEO catalysts by nearly a factor of 2 (6–14, 16, 18–24, 30). The above results indicated that the “performance ceiling” shown in Fig. 1A is successfully broken, with CO conversion and light-olefins selectivity both exceeding 80%.

Catalyst comparison

To further illustrate the versatility of GeAPO-18 zeotypes, we compared them with corresponding catalysts substituted with silicon (SAPO-18) and with magnesium (MgAPO-18). Those two zeotypes were prepared with similar methods

as that used for GeAPO-18 (supplementary materials) with a Si/Al ratio ranging from 0.011 to 0.029 and Mg/P ratio from 0.019 to 0.029, respectively. The resulting ZnCrO_x-MeAPO-18_a combinations were all tested in a wide temperature range of 410° to 480°C and space velocity between 1500 to 6000 ml/(g_{cat}·hour). The CO₂ selectivity appeared to be controlled thermodynamically (figs. S15 and S16). The red-orange-yellow region of the contour plots in Fig. 3, A to C, indicates light-olefins selectivity >80%, and black regions indicate selectivity <60%. The maps indicate that GeAPO-18 allowed the reaction to be operated over a wider temperature range, in which light-olefins selectivity and CO conversion simultaneously remained >80% with the (C₂=C₄)/(C₂^o-C₄^o) ratio > 15 (Fig. 3A and fig. S17). A wider operation temperature range is highly desirable for applications. By contrast, the red-orange-yellow operation zone was much smaller with the classical SAPO-18 zeotypes (Fig. 3B) and almost nonexistent with MgAPO-18 (Fig. 3C). All SAPO-18_a, MgAPO-18_a, and GeAPO-18_a samples examined under different conditions in this work are detailed in Fig. 3D. It reveals that the activity-selectivity trade-off boundary cannot be exceeded with SAPO-18 and MgAPO-18

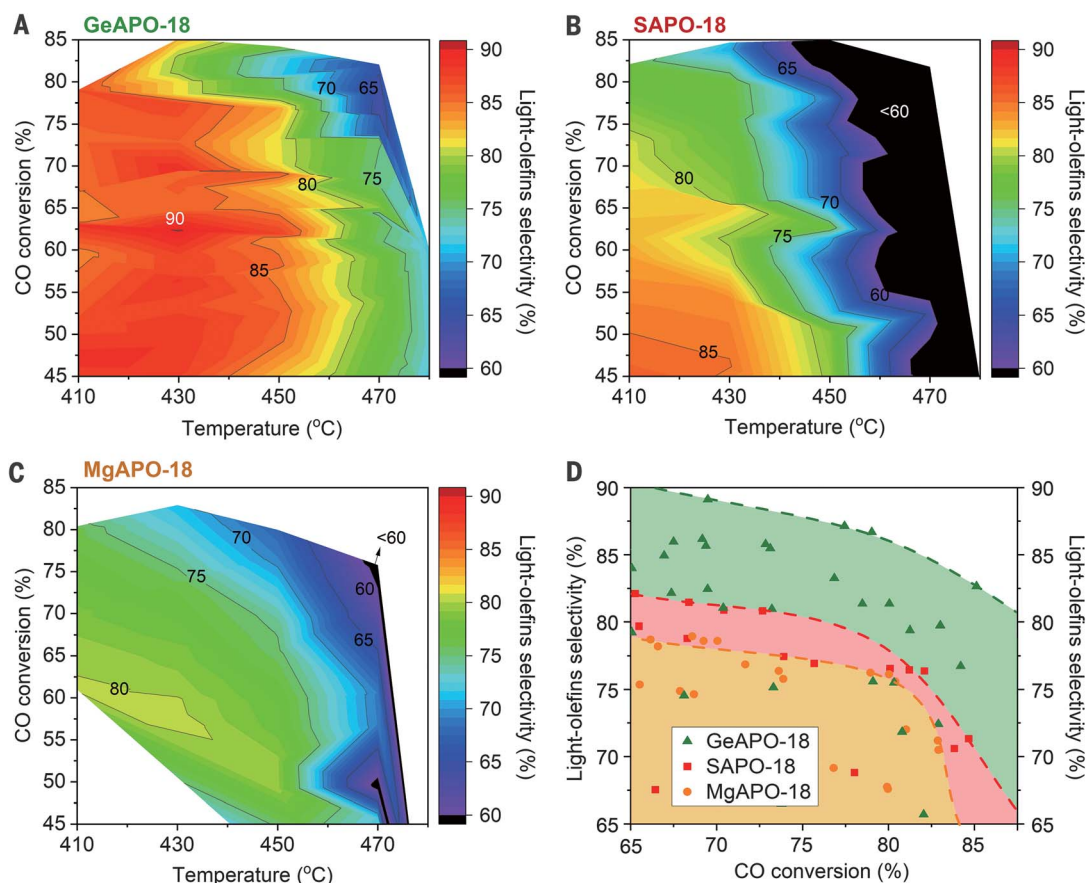


Fig. 3. Comparison of ZnCrO_x-MeAPO-18 in syngas conversion. Contour plots of catalytic performance. (A) GeAPO-18 with Ge/Al = 0.012 to 0.027. (B) SAPO-18 with Si/Al = 0.011 to 0.029. (C) MgAPO-18 with Mg/P = 0.019 to 0.029. (D) Selectivity of light olefins versus CO conversion, with the dashed lines being guides for the eye. Reaction condition: H₂/CO = 2.5, 410° to 480°C, 6 MPa, 1500 to 6000 ml/(g·hour).

by changing their acidity properties (varying Si/Al and Mg/P ratios) or the reaction conditions, but it can be breached by GeAPO-18.

Mechanistic studies

We compared GeAPO-18_{0.025}, SAPO-18_{0.016}, and MgAPO-18_{0.029} to understand the mechanism underlying the high yield of olefins over the ZnCrO_x-GeAPO-18 catalysts because all three catalysts have a similar density of the medium-strength acid sites (table S4) as well as comparable diffusion efficiency (fig. S18). The NH₃-TPD profiles in fig. S11 show that the desorption corresponding to the medium-strength acid sites peaks at 235°C for GeAPO-18_{0.025}, which is 50°C lower than that of SAPO-18_{0.016}. In comparison, MgAPO-18 shows a broad high-temperature desorption feature (between 300° and 413°C). The desorption temperatures reveal that the strength of the acidity decreased in the sequence of MgAPO-18 > SAPO-18 > GeAPO-18. These results were consistent with the experimentally measured differential adsorption heat of NH₃ (table S7) and NH₃ binding energy on different Brønsted acid sites calculated with density functional theory (DFT) (table S8 and figs. S8 and S19), which were, respectively, 143 kJ/mol and -1.24 eV for GeAPO-18_{0.025}, 156 kJ/mol, and -1.40 eV for SAPO-18_{0.016}, and 193 kJ/mol and -1.71 eV for MgAPO-18_{0.029}.

Because the light-olefins yield improved with decreasing strength of the zeotype acidity, we examined the effect of the acidity on the adsorption and activation of H₂ and olefins by using ethylene as a probe molecule (table S9 and figs. S8 and S20). The calculated mean ethylene adsorption energy was 0.56 ± 0.05 eV on the Ge-OH-Al Brønsted acid sites (Fig. 4A), which was noticeably weaker than that on Si-OH-Al and Mg-OH-P sites. This order correlated well with the NH₃ binding energy trend, even considering the variation of the energies on different T-sites of each zeotype (Fig. 4A), and was further confirmed with the ethylene-TPD experiments (Fig. 4B). The H-D exchange experiments revealed a weaker H₂ activation activity on GeAPO-18_{0.025} than that on SAPO-18_{0.016} and MgAPO-18_{0.029} (Fig. 4C) (31).

Both the weaker ethylene adsorption and H₂ activation over GeAPO-18 retard the secondary reactions of ethylene hydrogenation to ethane and its oligomerization to higher hydrocarbons C₃₊ (18). This is further demonstrated by ethylene conversion in the presence of hydrogen between 400° and 460°C over MeAPO-18 (Fig. 4D and fig. S21). GeAPO-18 exhibited lower activity for ethylene hydrogenation and oligomerization than those of the other two zeotypes. Although the presence of CO could significantly suppress the second-

ary reactions of olefins, GeAPO-18 still exhibited the weakest activity among the three zeotypes (fig. S22). The above results demonstrated that the Ge-OH-Al site can modulate the relative kinetics of C-C coupling of ketene to olefins and the subsequent over-hydrogenation and oligomerization of olefins and hence results in high selectivity for light olefins. The weaker oligomerization activity of Ge-OH-Al sites, as another advantage, led to a higher selectivity to more valuable ethylene and propylene compared with that of the optimized SAPO-18 under the same conditions (tables S10 and S11), which is of economic importance for industrial application. Therefore, CO conversion is improved by increasing the density of Ge-OH-Al sites and optimization of reaction conditions without degrading the light-olefins selectivity (Fig. 2A and fig. S23). An even higher activity can be expected if more Ge-OH-Al Brønsted acid sites could be generated as soon as a suitable preparative technique for enhanced heterometal substitution into zeotype frameworks becomes available (fig. S23).

Discussion

This study demonstrates that the key to breaking the activity and selectivity trade-off in CO hydrogenation to olefins lies in the disentanglement of the catalytically active sites for

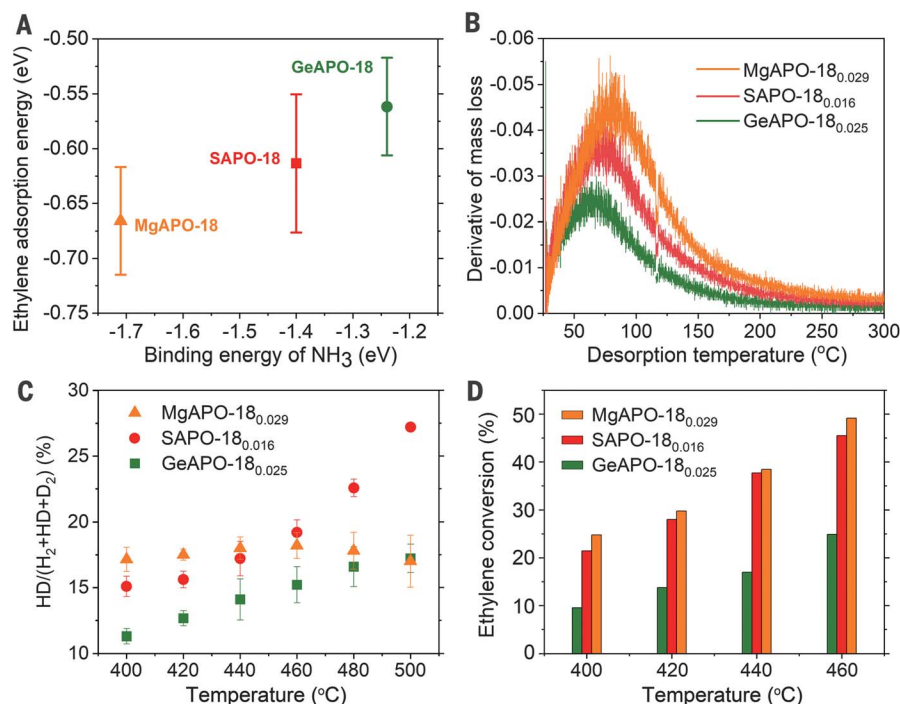


Fig. 4. Acidic strength of GeAPO-18 and its effects on the catalytic performance in comparison with SAPO-18 and MgAPO-18. (A) Calculated mean adsorption energy of ethylene versus the binding energy of NH₃. The error bars indicating the variation of adsorption energy between different T-sites of zeotypes (tables S8 and S9). (B) Ethylene-TPD measured by the derivative of the mass loss detected with intelligent gravimetric analysis. (C) H-D exchange activity represented by the relative HD/(H₂+HD+D₂) concentration. (D) Ethylene hydrogenation at 400° to 460°C, 4 MPa, 200 ml/min, feed 1.2% C₂H₄-66.6% H₂-32.2% Ar.

the target and side reactions. It demonstrates that incorporation of Ge-substituted AlPO-18 provides an effective strategy within the OXZEO catalyst concept. Its weaker Ge-OH-Al Brønsted acid sites are largely inactive for the secondary reactions of hydrogenation and oligomerization of olefins, and increasing the density of Brønsted acid sites does not strengthen the acidity. Therefore, one can create more active sites to catalyze the conversion of ketene intermediates to light olefins and thus shift the thermodynamically limiting step of syngas conversion to ketene. As a result, 85% CO conversion and 83% light-olefins selectivity can be simultaneously obtained, resulting in an unprecedented light-olefins yield of 48% under optimized conditions. Furthermore, metal substitution constitutes a general

strategy to mediate the strength and density of Brønsted acid sites in zeotype frameworks, which will enable the optimization of catalysts not only for CO but also CO₂ hydrogenation to value-added chemicals.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Supplementary Text
Figs. S1 to S23
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References

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Supplementary Materials for

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Feng Jiao *et al.*

Corresponding authors: Xiulian Pan, panxl@dicp.ac.cn; Xinhe Bao, xhbao@dicp.ac.cn

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The PDF file includes:

Materials and Methods
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Materials and Methods

Synthesis of ZnCrO_x

ZnCrO_x oxide was prepared following the previously reported procedure by co-precipitation method (6, 18, 24). Briefly, 14.56 g Zn(NO₃)₂·6H₂O, 5.6g Cr(NO₃)₃·9H₂O and 5.25g Al(NO₃)₃·9H₂O were dissolved in 100 mL distilled water. Subsequently, (NH₄)₂CO₃ aqueous solution was prepared as the precipitant. Co-precipitation was conducted at 70 °C. After being aged for 3 h, the precipitate was filtered and washed by distilled water, followed by drying at 110 °C and calcination at 500 °C for 1 h.

Synthesis of SAPO-18, GeAPO-18 and MgAPO-18

GeAPO-18, SAPO-18 and MgAPO-18 were synthesized via a hydrothermal method following the same procedure as the previous report (18). Typically, aluminium isopropoxide, 85% phosphoric acid, GeO₂ or 30% silica sol or Mg(NO₃)₂, *N,N*-diisopropylethylamine (C₈H₁₉N) were dissolved in distilled water with a molar composition of Al₂O₃: P₂O₅: GeO₂/SiO₂/MgO: C₈H₁₉N: H₂O = 0.9-1: 1: 0.04-0.07/0.02-0.05/0.04-0.11: 1.6-1.8: 42-46. The crystallization was allowed in a Teflon-lined autoclave at 160 °C for 7 days. The precipitate was washed with distilled water followed by drying overnight and calcination at 550-600 °C for 5 h. The resulting samples were denoted as MeAPO-18_a with ‘a’ representing the mass ratio of Ge/Al, Si/Al and Mg/P.

Synthesis of GeAPO-11 and GeAPO-5

GeAPO-11 and GeAPO-5 were synthesized via a hydrothermal method for comparison. Aluminium isopropoxide, 85% phosphoric acid, GeO₂, di-*n*-propilamina (DPA) were dissolved in distilled water with a molar composition of Al₂O₃: P₂O₅: GeO₂: DPA: H₂O = 1: 1: 0.5: 1.4: 50. The crystallization was allowed in a Teflon-lined autoclave at 200 °C for 1 day. The precipitate was washed with distilled water followed by drying overnight and calcination at 600 °C for 5 h. The resulting samples were denoted as GeAPO-11_{0.026}. Aluminium isopropoxide, aluminum sulfate, 85% phosphoric acid, GeO₂, *N,N*-diisopropylethylamine (C₈H₁₉N) and diethylamine (DEA) were dissolved in distilled water with a molar composition of Al₂O₃ (Al isopropoxide): Al₂O₃ (Al sulfate): P₂O₅: GeO₂: C₈H₁₉N: DEA: H₂O = 0.6: 0.4: 1: 0.08: 0.9: 1.1: 45. The crystallization was allowed in a Teflon-lined autoclave at 160 °C for 2 days. The precipitate was washed with distilled water followed by drying overnight and calcination at 600 °C for 5 h. The resulting samples were denoted as GeAPO-5_{0.031}. XRD and XRF results of these two zeotypes are displayed in fig. S1B and table S2. Two samples are combined with ZnCrO_x, respectively, for

syngas conversion under reaction condition of 430 °C, 6 MPa, 1500 mL/(g_{cat}·h). The results are displayed in table S12.

Catalyst characterization

X-Ray Diffraction (XRD) patterns were obtained using a PANalytical Empyrean-100 equipped with CuK α radiation source ($\lambda = 1.5418 \text{ \AA}$), operated at 40 mA and 40 kV.

The element composition of various zeotype samples were analyzed by a PANalytical Zetium X-Ray Fluorescence (XRF) spectrometer.

Water-droplet contact angle tests were conducted on the KRUSS contact angle goniometer.

Integrated differential phase contrast scanning transmission electron microscopy (iDPC-STEM) experiments were performed on the Titan Themis G3 environmental transmission electron microscope (ETEM, Thermo Scientific Company) at electron microscopy center of Dalian Institute of Chemical Physics, which worked at 300 kV with a spherical-aberration (Cs) corrector for parallel imaging (CEOS GmbH). The microscope was aligned prior to each experiment, using a cross-grating standard sample. Specimen searching and zone axis alignment were carried out under the TEM mode. After alignment of the crystal zone axis, the microscope was switched to the STEM mode with a convergence semi-angle of 9.8 mrad to acquire iDPC-STEM images. The probe current was $\sim 1 \text{ pA}$, and the dwell time is $2.5 \text{ }\mu\text{s/pixel}$. The iDPC-STEM images were acquired using a 4-quadrant DF4 detector with a high-pass filter applied on to reduce the low frequency information, and the collection angle was 2-10 mrad. Energy dispersive X-ray spectroscopy (EDS) elemental mappings over individual particles were collected using a JED-2300T, that performed on the transmission electron microscope (TEM, JEOL ARM300F). TEM worked at 300 kV with a probe spherical-aberration (Cs) corrector.

The morphology and crystal size of the GeAPO-18 were observed by a high-resolution scanning electron microscope (SEM) using JSM-7800F from Japan Electronics Co., Ltd. Experiments were performed under high vacuum conditions with electron beam voltages ranging from 200 V to 30 kV.

X-ray absorption spectroscopy (XAS) was carried out at the BL14W beamline of Shanghai Synchrotron Radiation Facility (SSRF), which was operated at 3.5 GeV in “top-up” mode with injection current of 220 mA. EXAFS data were recorded in fluorescence mode equipped with an Electro-Lytle detector. GeO₂ (99.9%, Macklin) was used as a reference.

Fourier Transform Infrared Spectroscopy (FT-IR) experiment was carried out on a BRUKER INVENIO S equipped with a MCT detector. About 30.0 mg sample was pressed into a wafer (diameter 13 mm) and placed inside a quartz cell. Then it was evacuated at 450 °C for 3-4 h to obtain $<10^{-4}$ Pa. After cooling down to room temperature, the spectra from 4000 to 1000 cm^{-1} were collected by averaging 32 scans collected at 4 cm^{-1} resolutions. Thereafter, NH_3 adsorption was allowed at room temperature for 30 min and subsequently degassed by evacuation at 100 °C. All spectra were collected under room temperature.

Temperature-Programmed-Desorption of NH_3 (NH_3 -TPD) was conducted on a Micromeritics AutoChem 2920 instrument equipped with a thermal conductivity detector (TCD) detector. Typically, 100 mg sample was heated under a flowing Ar at 450 °C for 1.5 h. After cooling down to 100 °C, the sample was exposed to 5% NH_3/He for 30 min. Thereafter the sample was first swept by Ar at the same temperature for 30 min and further swept using He until a stable baseline of TCD signal was obtained. Subsequently, the TCD signal was recorded with the temperature increased from 100 to 800 °C at a heating rate of 10 °C/min.

Microcalorimetric measurements of ammonia were performed using a BT 2.15 heat flux calorimeter (Cetram, France) following previously reported procedures (18). The calorimeter was connected to a gas handling and volumetric system that employed a MKS 698A Baratron capacitance manometer for precision pressure measurement ($\pm 1.33 \times 10^{-2}$ Pa). Before adsorption, the samples were evacuated in a quartz cell at 450 °C for 4 h. The adsorption experiments were carried out at 150 °C.

Ethylene-temperature programmed desorption (TPD) was conducted on an Intelligent Gravimetric Analyzer (IGA100, Hiden Analytical). Typically, 150 mg sample was pretreated at 300 °C in vacuum for 4 h. After cooling down to room temperature, the sample was exposed to pure C_2H_4 at 1 bar for 2 h. Subsequently, the sample was swept by He for 5 h at room temperature until a stable baseline was obtained. The mass loss was recorded with the temperature increasing from RT to 300 °C at a heating rate of 5 °C/min.

The uptake rates of ethane inside zeotypes were conducted on IGA100. Typically, 30 mg zeotypes (40-60 mesh) was degassed at 300 °C for 4h and cooled down to 30 °C until a constant weight baseline. Thereafter, ethane was introduced to the chamber and the adsorption kinetics curves were recorded at 30 °C and 5 mbar. The diffusion coefficient was analyzed according to the

Fick's second law of diffusion following our previous study (22). The shape of zeotype crystal was simplified to a spherical structure model assuming the same diffusion length of 100 nm.

Catalytic reaction

Syngas reaction was carried out in a high throughput 16-channel fixed bed reactors (Avantium Flowrence T1701) with a 2 mm inner diameter. ZnCrO_x and zeotypes were mixed with a mass ratio of 2:1 and a granule size of 40-60 mesh. 5% Ar was included in the feed syngas ($\text{H}_2/\text{CO} = 2.5$) as the internal standard for online gas chromatography (GC) analysis (Agilent 7890B). Hydrocarbons and oxygen-containing compounds were analyzed by two flame ionization detectors (FID) connected with HP-FFAP and HP-AL/S capillary columns. CO , CO_2 , C_2H_4 , and Ar were analyzed by a TCD connected with Porapak Q and 5 Å molecular sieves packed columns. C_2H_4 was taken as a reference bridge between FID and TCD.

CO conversion was calculated on a carbon atom basis, i.e.

$$\text{Conv}_{\text{CO}} = \frac{\text{CO}_{\text{inlet}} - \text{CO}_{\text{outlet}}}{\text{CO}_{\text{inlet}}} \times 100\% \quad (1)$$

Where CO_{inlet} and $\text{CO}_{\text{outlet}}$ represent moles of CO at the inlet and outlet, respectively.

CO_2 selectivity based on C atom (Sel_{CO_2}) was calculated according to:

$$\text{Sel}_{\text{CO}_2} = \frac{\text{CO}_{2\text{outlet}}}{\text{CO}_{\text{inlet}} - \text{CO}_{\text{outlet}}} \times 100\% \quad (2)$$

Where $\text{CO}_{2\text{outlet}}$ denotes moles of CO_2 at the outlet.

The selectivity of individual hydrocarbon C_nH_m ($\text{Sel}_{\text{C}_n\text{H}_m}$) among all hydrocarbon products based on C atom (CO_2 excluded) was obtained according to:

$$\text{Sel}_{\text{C}_n\text{H}_m} = \frac{n_{\text{C}_n\text{H}_m\text{outlet}}}{\sum_i^n n_{\text{C}_n\text{H}_m\text{outlet}}} \times 100\% \quad (3)$$

Note that the yield of light olefins was calculated by multiplying CO conversion with the selectivity of light olefins among all carbon-containing products including CO_2 .

To study the stability and Zn migration effect, we carried out experiment following the similar procedures as our previous study (24). First, $\text{GeAPO-18}_{0.025}$ granules of 20-40 mesh were physically mixed with ZnCrO_x of 60-80 mesh, and subjected to syngas conversion under 430 °C, 6 MPa, 1500 mL/(g_{cat}·h) for 100 h. After reaction, the used GeAPO-18 granules were carefully separated and characterized.

H-D exchange experiments were conducted in a fixed bed reactor with the effluents monitored by an on-line mass spectroscopy (MS). Typically, 100 mg sample was loaded in an inert

quartz tube and pretreated under a flowing Ar at 450 °C for 5 h. After cooling down to room temperature, the sample was swept using 2.5 mL/min D₂ and 25 mL/min 10%H₂/Ar for 40 min until a stable baseline was obtained. Subsequently, the MS signals were recorded while the temperature was increased at a ramp rate of 10 °C/min.

Ethylene/H₂ conversion model experiments were performed in a fixed-bed stainless steel reactor equipped with a quartz lining. Typically, 250 mg sample (20-40 mesh) was diluted by 200 mg inert quartz sand. The catalyst was swept in N₂ from room temperature to 400 °C, then a pre-mixed gas, composed of 1.2% C₂H₄, 66.6% H₂ and 32.2% Ar, was introduced as the feed at 100-200 mL/min and the reaction was allowed at 4 MPa and 400 - 460 °C. Products were analyzed by an online GC (Shimadzu GC-2014), equipped with an FID and a TCD. To clarify the effect of CO in this model experiments, 10% CO balanced with Ar was simultaneously introduced into the reactor at a flow rate of 40 mL/min.

C₂H₄ conversion was calculated on a carbon atom basis, i.e.

$$\text{Conv}_{\text{C}_2\text{H}_4} = \frac{C_{2\text{H}_4\text{inlet}} - C_{2\text{H}_4\text{outlet}}}{C_{2\text{H}_4\text{inlet}}} \times 100\% \quad (4)$$

Where $C_{2\text{H}_4\text{inlet}}$ and $C_{2\text{H}_4\text{outlet}}$ represent moles of C₂H₄ at the inlet and outlet, respectively.

DFT calculation

All energetics and structural relaxations were calculated using density functional theory (DFT), as implemented in the Vienna *ab initio* simulation package (VASP). In addition, the projector augmented-wave (PAW) scheme and generalized gradient approximation (GGA) method at the level of the revised Perdew–Burke–Ernzerhof (rPBE) functional were used. The convergence criteria of energy and force were set to 10⁻⁴ eV and 0.05 eV·Å⁻¹ in structural optimizations. Kinetic energy cutoff was specified to 400 eV, and sampled the Brillouin zone with the Γ point.

An AlPO-18 unit cell was used with $a = 13.71 \text{ \AA}$, $b = 12.73 \text{ \AA}$, $c = 18.57 \text{ \AA}$.

The adsorption energy of ammonia and ethylene is calculated as follows:

$$E_{\text{AD-NH}_3} = E_{\text{total}} - E_{\text{ZEO}} - E_{\text{NH}_3}$$

$$E_{\text{AD-C}_2\text{H}_4} = E_{\text{total}} - E_{\text{ZEO}} - E_{\text{C}_2\text{H}_4}$$

E_{total} is the energy of zeolite after adsorbing gas molecules, and E_{ZEO} is the energy of individual zeolite. The adsorption models are optimized under the condition of all atoms relaxation.

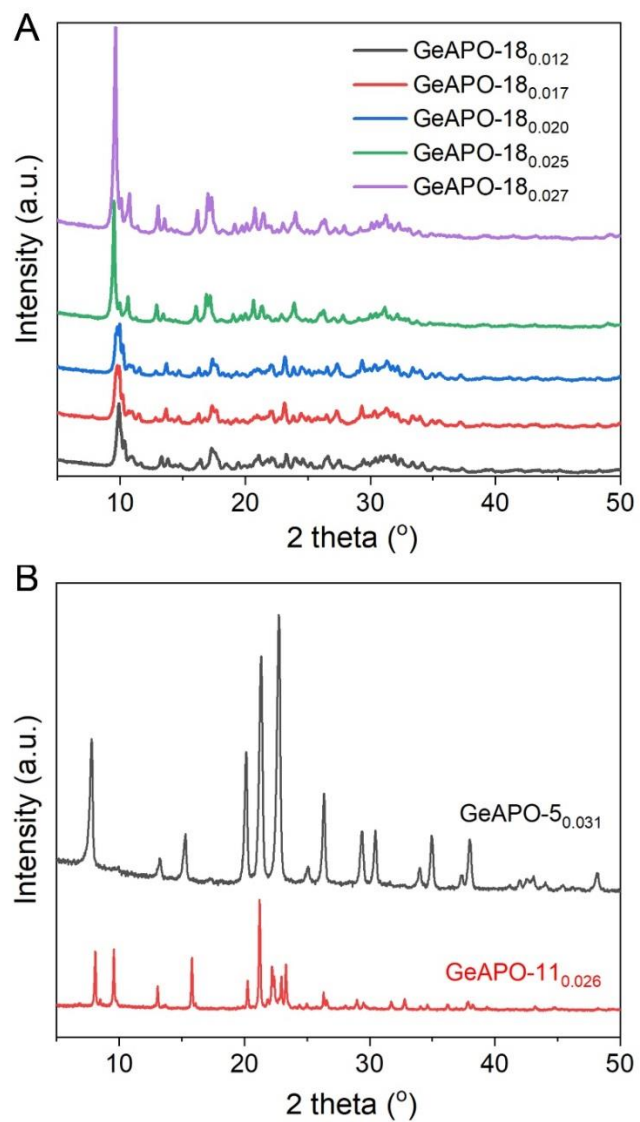


Fig. S1. XRD patterns of (A) GeAPO-18_a, (B) GeAPO-11_{0.026} and GeAPO-5_{0.031} with the subscript denoting the mass ratio of Ge/Al measured by XRF.

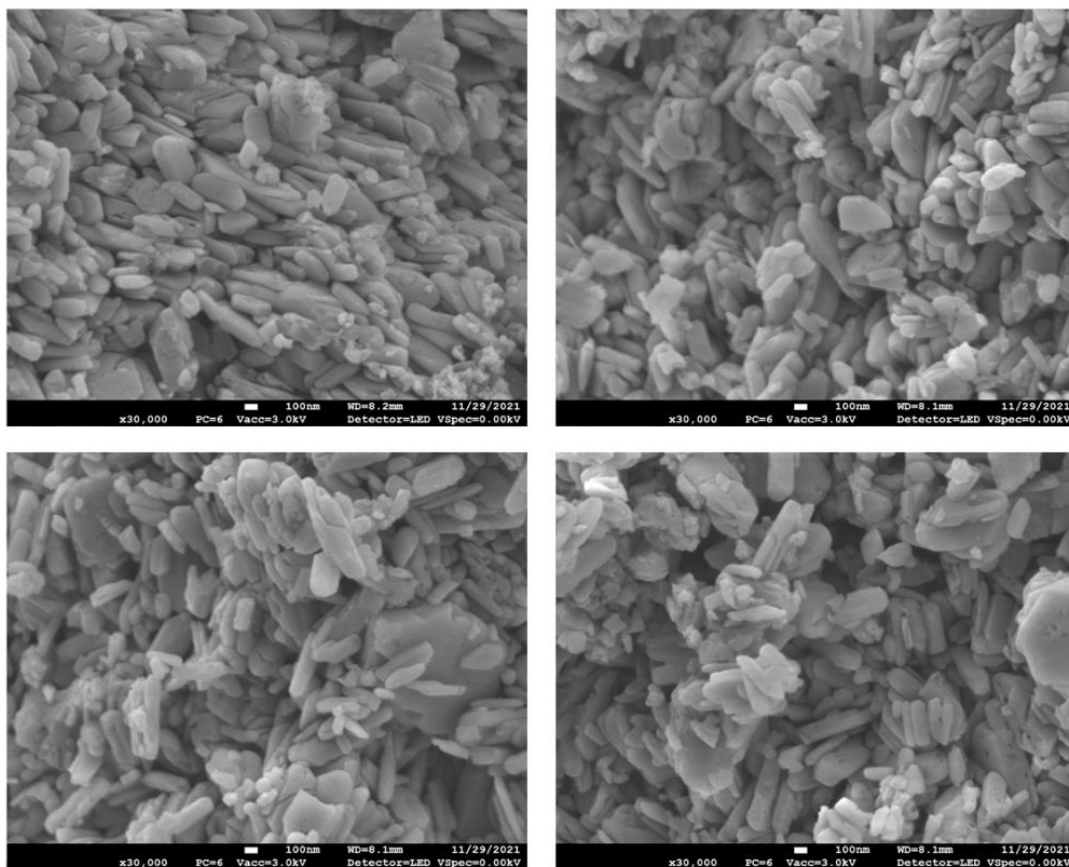


Fig. S2. SEM images of the randomly scanned regions of GeAPO-18_{0.025}.

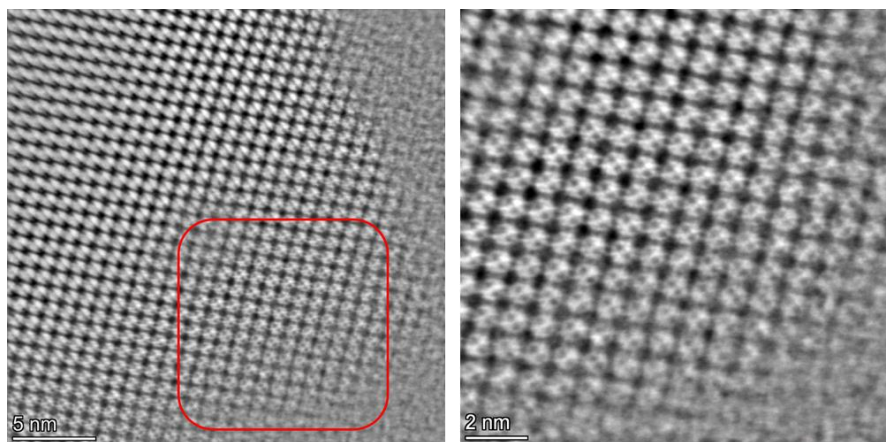


Fig. S3. Integrated differential phase contrast scanning transmission electron microscopy (iDPC-STEM) images of GeAPO-18_{0.025}, which shows a characteristic AEI structure.

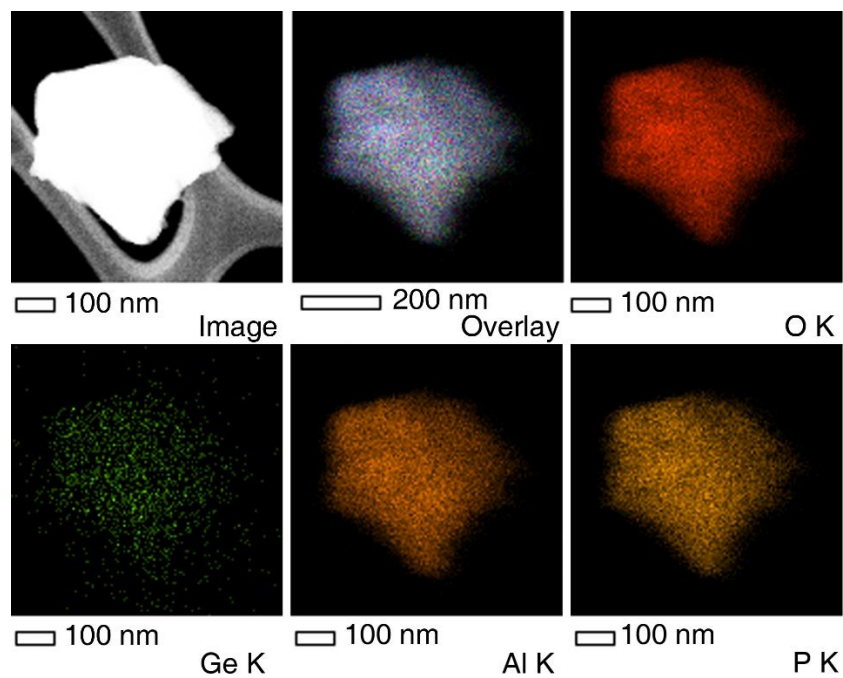


Fig. S4. Element mapping of GeAPO-18_{0.025}, which shows homogeneously distributed Ge species.

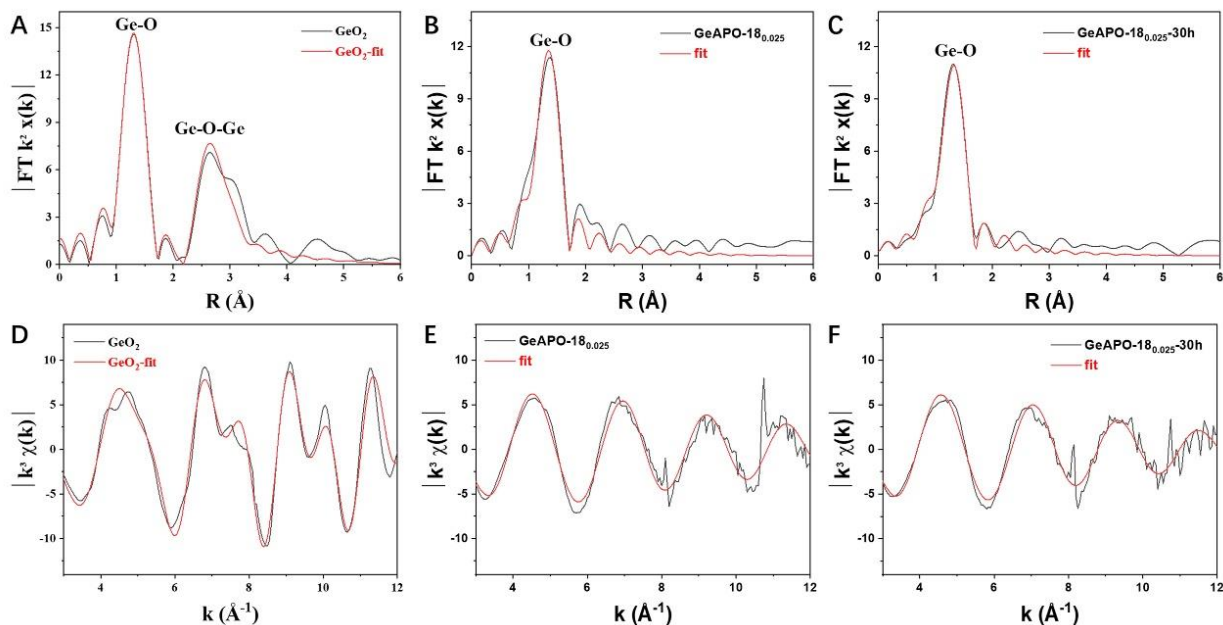


Fig. S5. X-ray Absorption Spectra of the Ge K-edge in the R-space and k-space. (A, D) GeO_2 reference; **(B, E)** Fresh $\text{GeAPO-18}_{0.025}$; **(C, F)** $\text{GeAPO-18}_{0.025}$ in combination with ZnCrO_x after syngas reaction for 30 h. Fig. S5B shows only the first shell Ge-O at distance of 1.1 Å and no second shell corresponding to Ge-O-Ge as in GeO_2 reference (fig. S5A). It confirms that the germanium species are dispersed as single sites throughout GeAPO-18 . EXAFS analysis indicates that Ge is fourfold coordinated with the O atom (table S3), which is characteristic for the framework Ge, revealing the location of Ge within the framework of GeAPO-18 . The coordination structure of Ge does not change even after 30 h reaction (fig. S5C), indicating a stable structure.

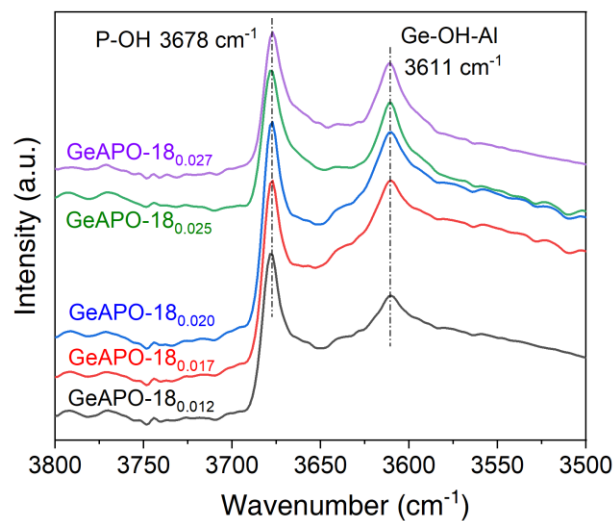


Fig. S6. IR spectra of the hydroxyl stretching vibration of GeAPO-18_a samples. All samples exhibit a strong absorption band at 3611 cm⁻¹. It is characteristics of the Brønsted acid sites O-H vibration associated with Ge-OH-Al. In addition, the vibration corresponding to P-OH (3678 cm⁻¹) is also observed (16, 18).

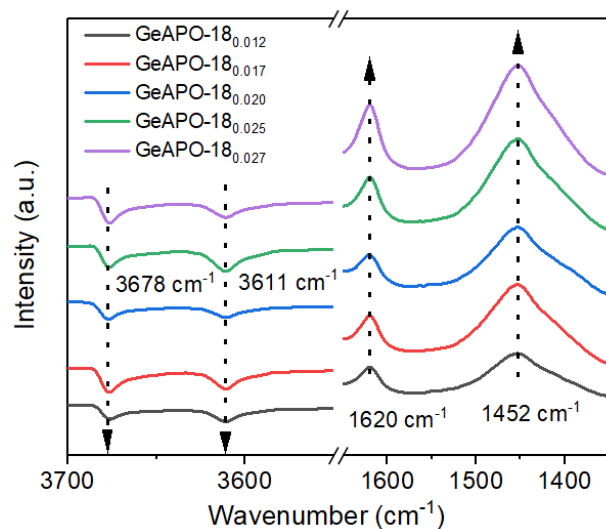


Fig. S7. Differential IR spectra of NH_3 -adsorbed GeAPO-18 by referring to that of the corresponding pristine GeAPO-18. The negative bands 3611 and 3678 cm^{-1} in the differential spectra correspond to consumed Brønsted acid sites and P-OH sites upon NH_3 adsorption, respectively. The bands 1452 cm^{-1} is attributed to the NH_3 -adsorbed Brønsted acid sites or P-OH, and 1620 cm^{-1} is attributed to Lewis acid sites (16, 18). These data further validate the successful substitution of P by Ge atoms within the framework of AlPO-18, and thus generating Brønsted acid sites.

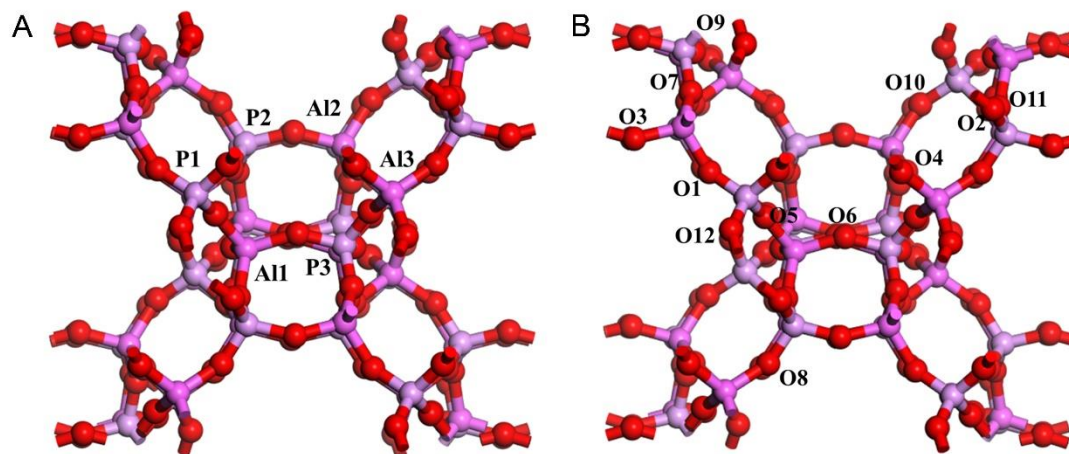


Fig. S8. Structure scheme of AEI. (A) Al, P sites. **(B)** O sites with the pink, red and purple balls representing Al, O and P, respectively.

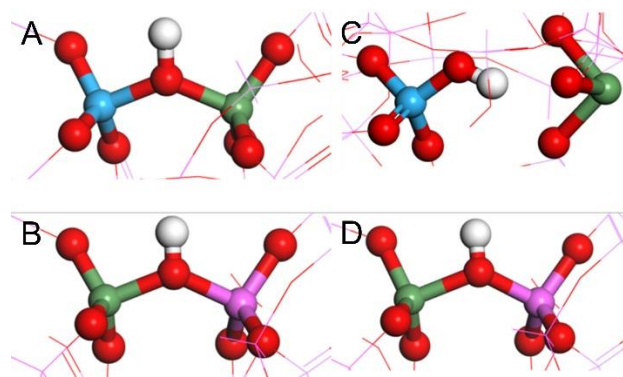


Fig. S9. DFT calculation of GeAPO-18 structural stability. (A) The initial structure model with Ge replacing the Al site (T1) of AEI structure (fig. S8). (B) The initial structure model with Ge replacing the P site (T1). (C) The final structure after relaxation, i.e. Ge-O bond being dissociated, indicating the unstable structure of Ge-OH-P. (D) The final structure Ge-OH-Al after relaxation, which retains the same as the initial one, demonstrating a stable structure of Ge-OH-Al. The green, pink, blue, red, and white balls represent the Ge, Al, P, O and H atoms, respectively. Therefore, Ge species most likely replace the P site in the aluminophosphate framework.

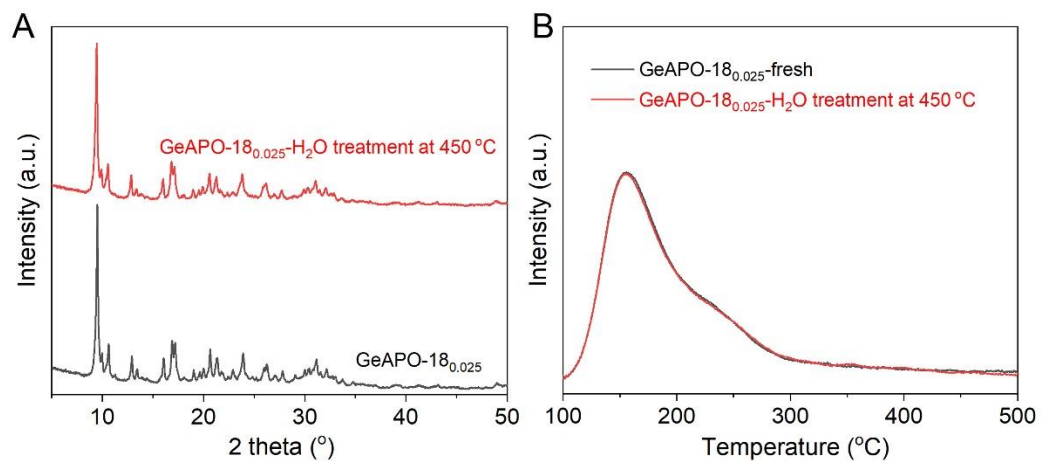


Fig. S10. (A) XRD patterns and (B) NH₃-TPD profiles for GeAPO-18_{0.025} before and after hydrothermal treatment in flowing N₂ containing 10% mol H₂O at 450 °C for 4 h.

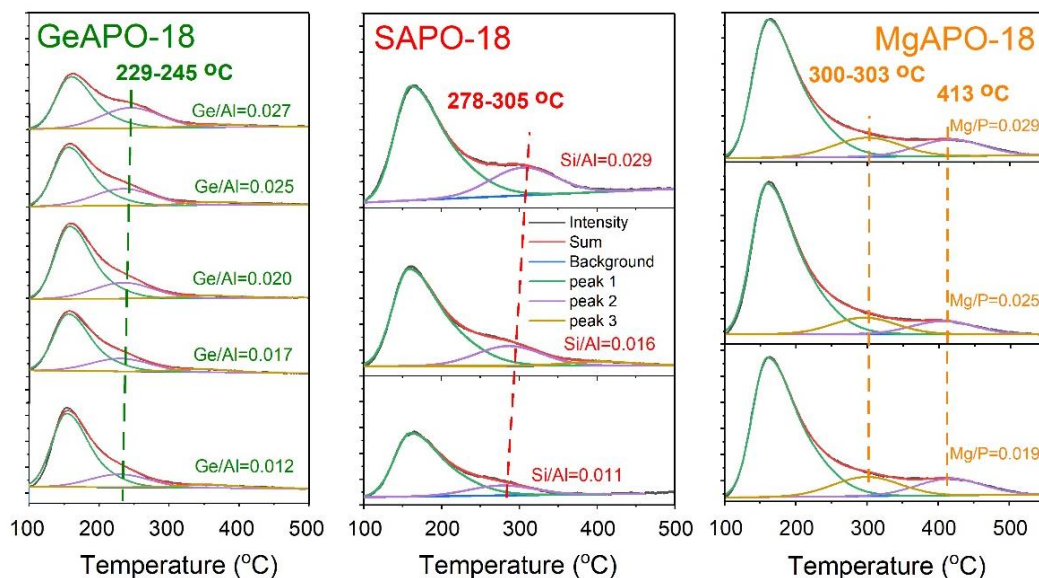


Fig. S11. NH₃-TPD profiles for GeAPO-18, SAPO-18 and MgAPO-18 with varying Ge/Al, Si/Al and Mg/P ratios. Deconvolution of the profiles indicates that the peak maximum being located below 200 °C is attributed to NH₃ desorption from the weak acid sites while that in the range from 229-413 °C to the medium strength acid sites. Their densities can be estimated according to the integrated area and are listed in table S4. The NH₃ desorption temperature reflects the acidity strength.

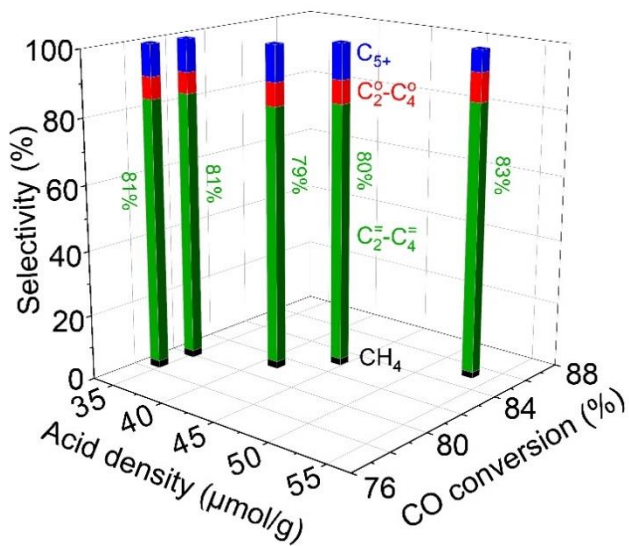


Fig. S12. Catalytic performance as a function of the medium strength acid density of GeAPO-18. The density of the medium strength acid sites was estimated according to the integrated desorption peaks of NH_3 -TPD in fig. S11 and summarized in table S4. Reaction condition: $\text{ZnCrO}_x/\text{GeAPO-18}$ mass ratio = 2, 430 °C, H_2/CO = 2.5, 6 MPa and 1500 mL/(g_{cat}·h).

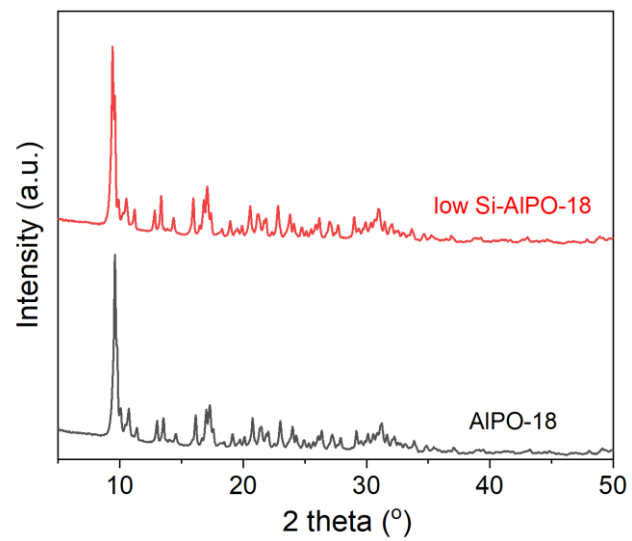


Fig. S13. XRD patterns of AlPO-18 and low Si-AlPO-18.

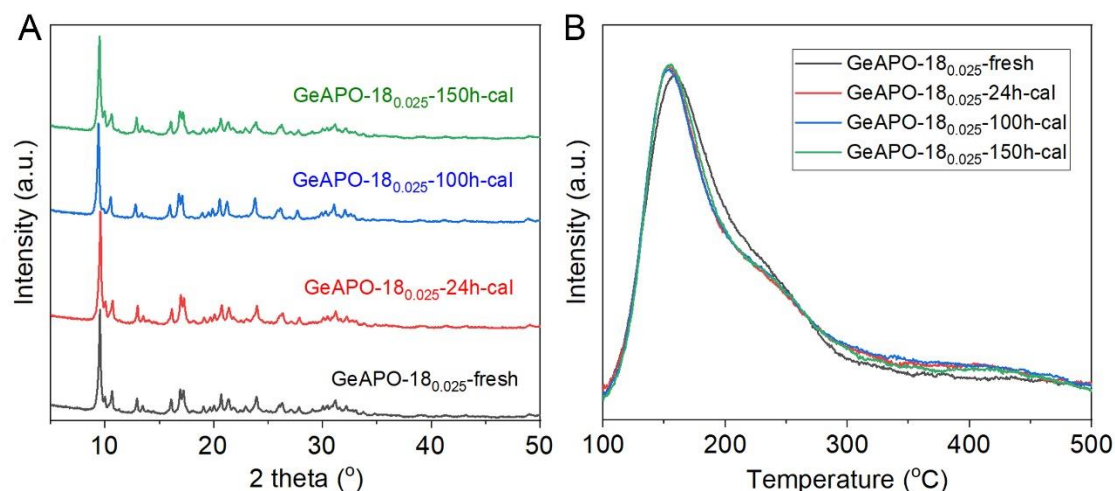


Fig. S14. (A) XRD patterns and (B) NH₃-TPD for GeAPO-18_{0.025} before and after syngas reaction for 24, 100 and 150 h. The crystal structure of the GeAPO-18 remains intact after 150 h reaction test. Prior to NH₃-TPD, the sample was subjected to calcination in air at 600 °C to eliminate the influence from any carbon species. Fig. S14B reveals that there is slight change for the profile of the sample after 24 h reaction in comparison to the fresh sample. However, the NH₃-TPD profiles are almost identical for the samples after 24, 100 and 150 h, demonstrating stable Brønsted acidity during the reaction, consistent with the stable performance in Fig. 2B. Therefore, the Ge species remain in the framework of GeAPO-18 after reaction. ICP-OES analysis of the used zeotype (table S6) indicates a negligible migration of Zn species.

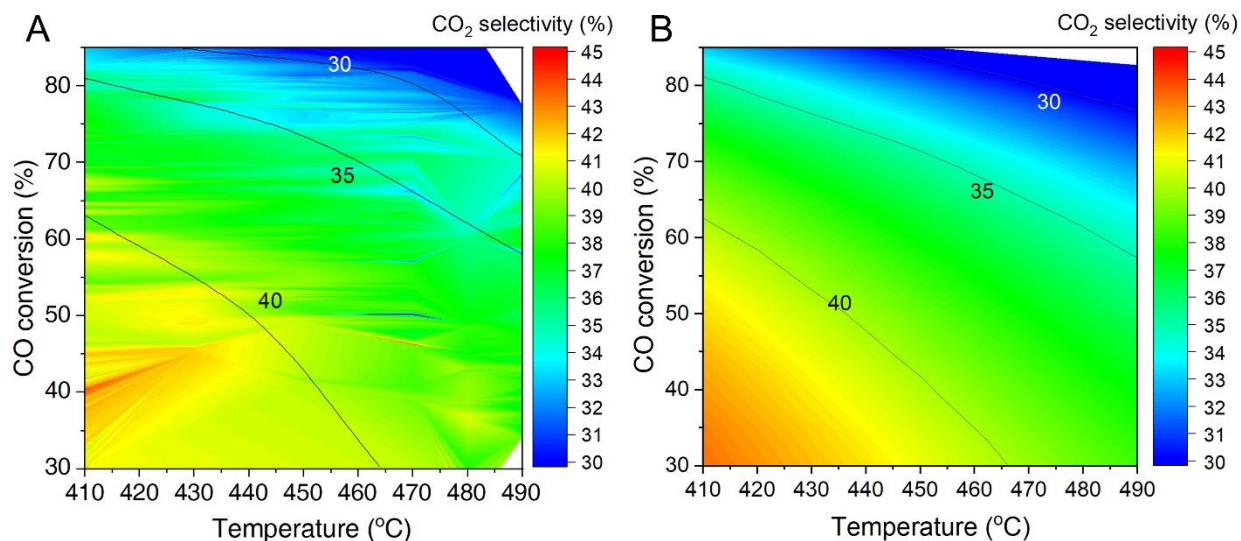
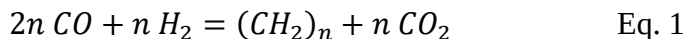


Fig. S15. Contour plots of CO₂ concentration as a function of temperature and CO conversion. (A) Experimental data corresponding to Fig. 3. (B) CO₂ concentration at the thermodynamic equilibrium calculated using HSC 9 assuming C₂H₄, CO₂ and H₂O as products under CO conversion between 30% and 90%. Reaction condition: H₂/CO = 2.5, 6 MPa, temperature 410-490 °C. The two contours are very similar, suggesting that CO₂ selectivity is largely controlled by thermodynamics and it will decrease further with the increasing CO conversion.



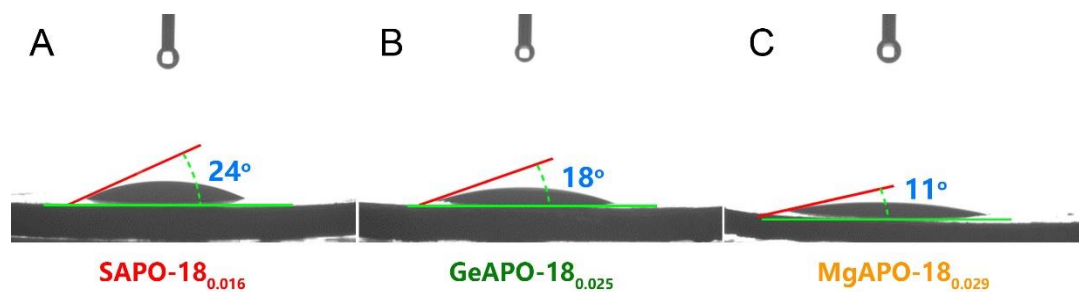


Fig. S16. Water-droplet contact angle. (A) SAPO-18_{0.016} (B) GeAPO-18_{0.025} and (C) MgAPO-18_{0.029}. The three zeotypes all exhibit a contact angle less than 30°, indicating a similar hydrophilic property.

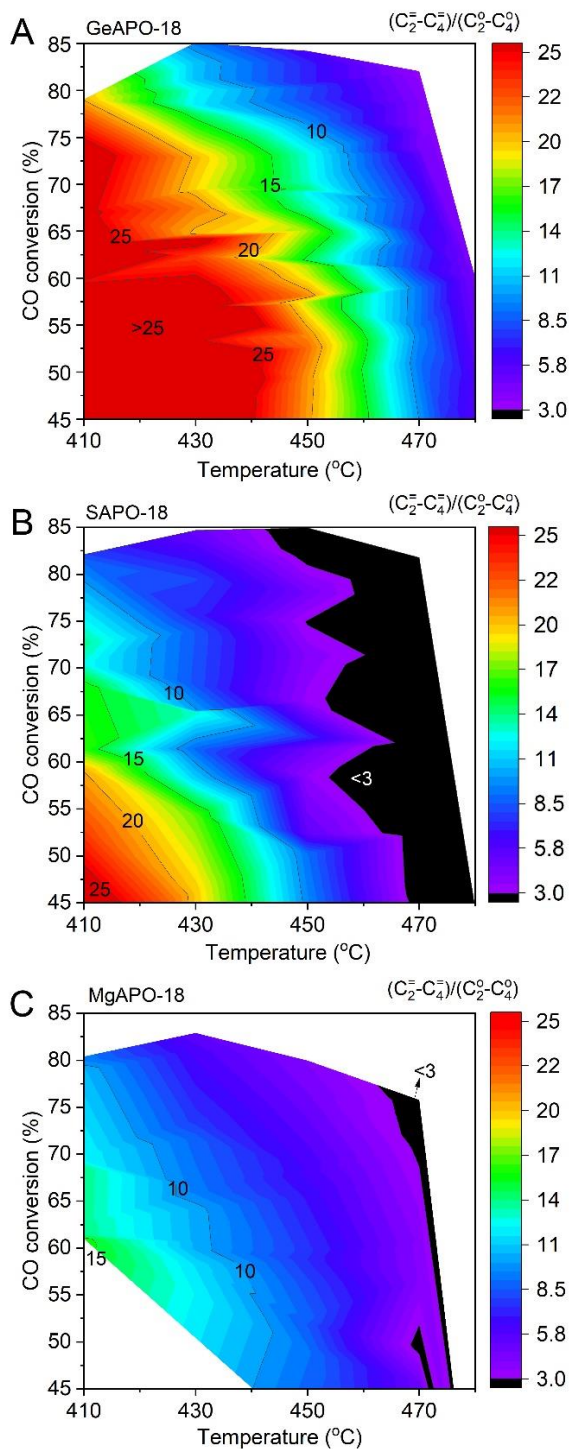


Fig. S17. Contour plots of the ratio of light olefins to paraffins as functions of CO conversion and reaction temperature. (A) ZnCrO_x-GeAPO-18; (B) ZnCrO_x-SAPO-18; (C) ZnCrO_x-MgAPO-18. The red-orange-yellow-green region represents $(C_2^= - C_4^=)/(C_2^o - C_4^o) > 15$ and the black region represents $(C_2^= - C_4^=)/(C_2^o - C_4^o) < 3$.

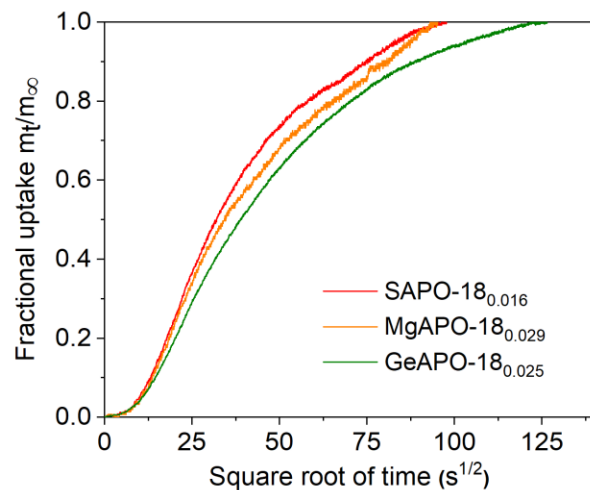


Fig. S18. Diffusion of ethane in GeAPO-18_{0.025}, SAPO-18_{0.016} and MgAPO-18_{0.029}, analyzed by IGA at 30 °C. The diffusion coefficients of ethane were estimated to be $2.9 \times 10^{-18} \text{ m}^2/\text{s}$ for SAPO-18_{0.016}, $2.3 \times 10^{-18} \text{ m}^2/\text{s}$ for MgAPO-18_{0.029} and $2.1 \times 10^{-18} \text{ m}^2/\text{s}$ for GeAPO-18_{0.025}, assuming the same diffusion length (100 nm) taking the crystal size into consideration. Therefore, the diffusion efficiency of ethane was similar among the three zeotypes.

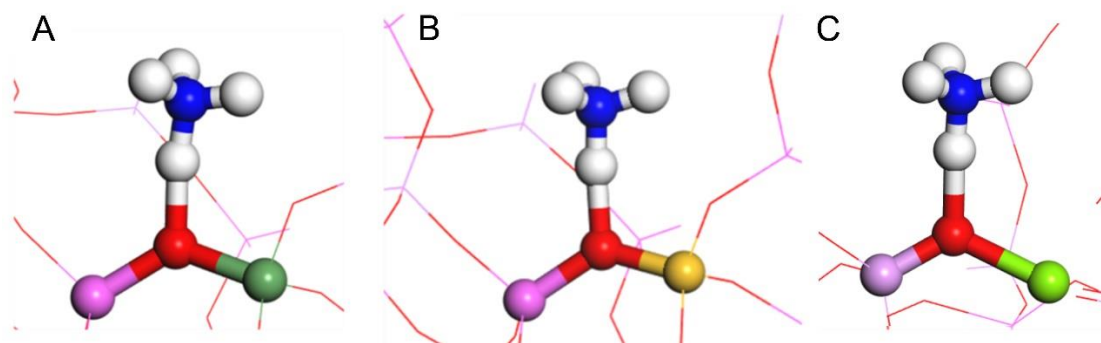


Fig. S19. The optimized structures of NH_3 binding on the Brønsted acid sites of (A) Ge-OH-Al (T3); (B) Si-OH-Al (T3) and (C) Mg-OH-P (T1) with the olive, yellow, green, pink, red, blue and white balls representing Ge, Si, Mg, Al(P), O, N and H atoms, respectively.

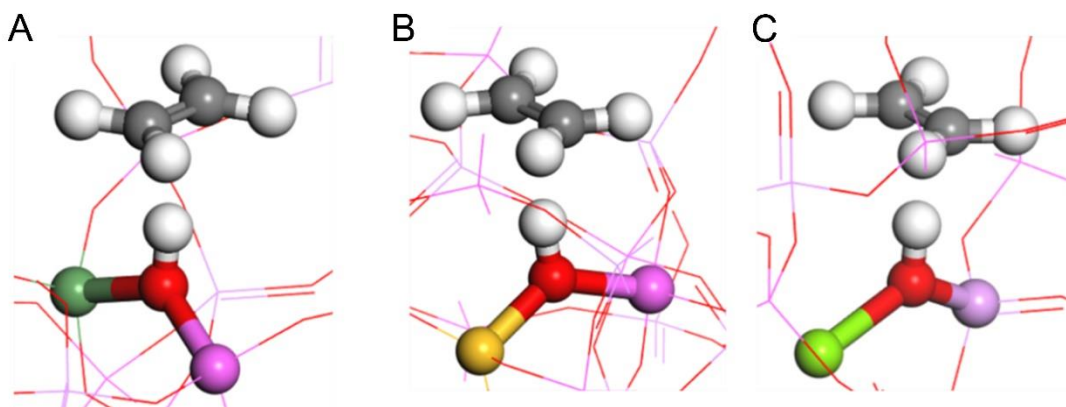


Fig. S20. The optimized structures of ethylene binding on the Brønsted acid sites of (A) Ge-OH-Al; (B) Si-OH-Al and (C) Mg-OH-P with the olive, yellow, green, pink, red, grey and white balls representing Ge, Si, Mg, Al(P), O, C and H atoms, respectively.

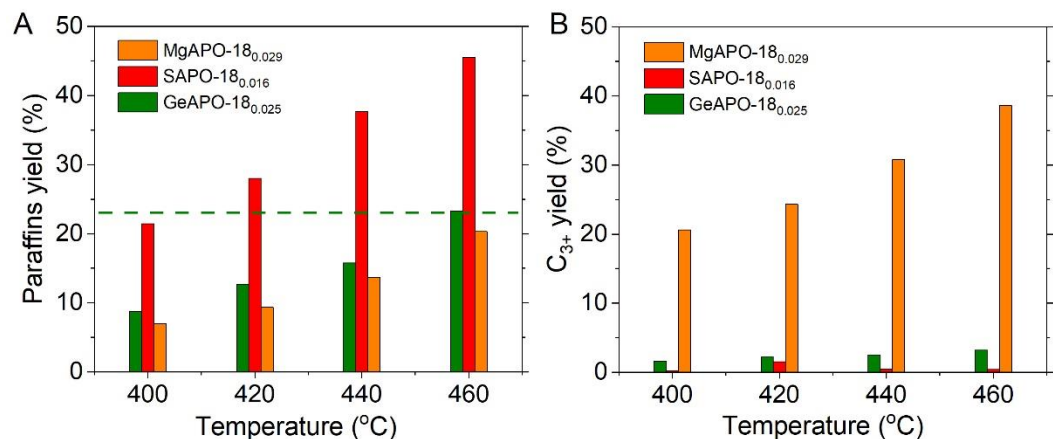


Fig. S21. Hydrogenation and oligomerization activity during ethylene conversion in H₂. (A) Paraffins yield and **(B)** C₃₊ yield. Reaction condition: 250 mg sample diluted by 200 mg quartz sands, 400 - 460 °C, 4 MPa, 200 mL/min feed composed of 1.2% C₂H₄, 66.6% H₂ and 32.2% Ar. Note that the hydrogen activity is reflected by the yield of paraffins and the oligomerization activity by the yield of hydrocarbons containing 3 and more than 3 carbon atoms.

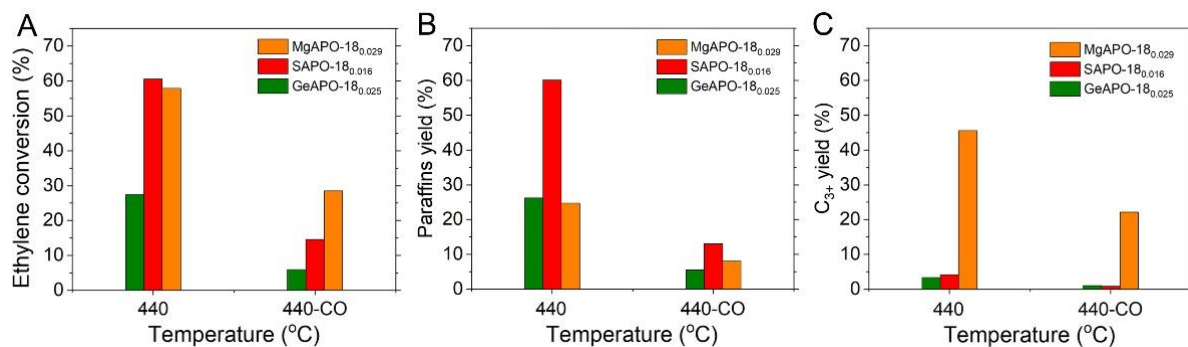


Fig. S22. Effect of CO on the ethylene/H₂ model reaction. (A) Ethylene conversion; **(B)** Paraffins yield and **(C)** C₃₊ yield. Reaction condition: 250 mg sample diluted by 200 mg quartz sand, 440 °C, 4 MPa, 100 mL/min gas mixture composed of 1.2% C₂H₄-66.6% H₂-32.2% Ar with/without CO (40 mL/min, 10% CO-90% Ar). The hydrogenation activity is reflected by the yield of paraffins and the oligomerization activity by the yield of hydrocarbons containing 3 and more than 3 carbon atoms.

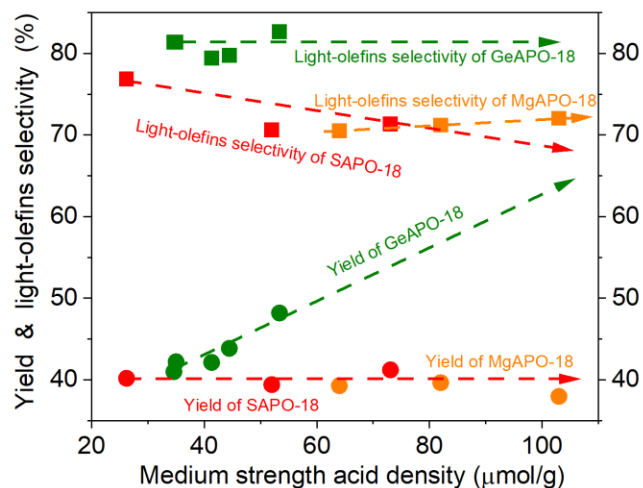


Fig. S23. The yield and selectivity of light olefins as a function of medium strength acid density of GeAPO-18 and SAPO-18. Reaction condition: $\text{H}_2/\text{CO} = 2.5$, 430 °C, 6 MPa, 1500 mL/(g_{cat}·h). It shows that the $\text{C}_2^+=\text{C}_4^+$ selectivity over GeAPO-18 keeps almost unchanged (green line) whereas it decreases (red line) with the increasing density of medium strength acid sites over SAPO-18. As a result, the yield of light olefins increases almost linearly with the acid site density over GeAPO-18. The extrapolated line (dashed green line) predicts that the yield could go beyond 60% at a density of 90 μmol/g.

Table S1. Summary of the reported yield of light olefins in direct syngas conversion.

Catalyst	CO conversion, %	C ₂ =-C ₄ = selectivity, %		C ₂ =-C ₄ = yield, %	Ref.
		CO ₂ -excluded	CO ₂ -included		
Fe/CNF ^a	88	52	30	26	(8)
FeMn@Si ^b	56	~27	~24	~14	(9)
Na-FeC _x /s-S1-OH ^c	83	29	15	13	(10)
Mn-promoted Co ₂ C ^d	32	61	32	10	(11)
CoMn-Na ₂ S ^e	0.8	54	54	0.4	(12)
CoMnC/PDVB ^f	64	71	38	24	(13)
ZnCrO _x -SAPO-34 ^g	17	80	47	8	(6)
ZnCrO _x -SAPO-18 ^h	49	83	42	21	(16)
ZnCrO _x -SAPO-34 ⁱ	60	75	45	27	(24)
ZnGaO _x -SAPO-34 ^j	50	75	45	23	(21)
ZnCrO _x -GeAPO-18 _{0.027} ^k	85	83	57	48	This work
ZnCrO _x -GeAPO-18 _{0.027} ^l	70	89	57	40	This work

^a H₂/CO = 1, 340 °C, 2 MPa, 1500 mL/(g·h)^b H₂/CO = 2, 320 °C, 2 MPa, 4000 mL/(g·h) estimated from Fig. 1B of ref. (9)^c H₂/CO = 1, 260 °C, 2 MPa, 600 mL/(g·h)^d H₂/CO = 2, 250 °C, 0.1 MPa, 2000 mL/(g·h)^e H₂/CO = 2, 240 °C, 0.1 MPa, 27000 mL/(g·h)^f H₂/CO = 2, 250 °C, 0.1 MPa, 900 mL/(g·h)^g H₂/CO = 2.5, 400 °C, 2.5 MPa, 7714 mL/(g·h)^h H₂/CO = 1, 390 °C, 10 MPa, 3600 mL/(g·h)ⁱ H₂/CO = 2.5, 400 °C, 4 MPa, 5000 mL/(g·h)^j H₂/CO = 2.5, 400 °C, 4 MPa, 1600 mL/(g·h)^k H₂/CO = 2.5, 430 °C, 6 MPa, 1500 mL/(g·h)^l H₂/CO = 2.5, 430 °C, 6 MPa, 4500 mL/(g·h)

Table S2. Elementary composition of GeAPO-18, SAPO-18 and MgAPO-18. ^a

Sample	Si, Ge or Mg wt%	Al wt%	P wt%	O w%	Ge/Al, Si/Al or Mg/P ratio
GeAPO-18 _{0.012}	0.28	23.00	24.50	52.22	0.012
GeAPO-18 _{0.017}	0.39	23.29	24.18	52.15	0.017
GeAPO-18 _{0.020}	0.46	23.22	24.16	52.15	0.020
GeAPO-18 _{0.025}	0.59	23.22	24.09	52.06	0.025
GeAPO-18 _{0.027}	0.64	23.43	23.91	52.01	0.027
SAPO-18 _{0.011}	0.26	23.84	23.73	52.15	0.011
SAPO-18 _{0.016}	0.39	23.60	23.81	52.20	0.016
SAPO-18 _{0.029}	0.70	23.70	23.44	52.15	0.029
MgAPO-18 _{0.019}	0.46	23.16	24.21	52.17	0.019
MgAPO-18 _{0.025}	0.62	22.77	24.42	52.19	0.025
MgAPO-18 _{0.029}	0.70	22.85	24.30	52.16	0.029
GeAPO-11 _{0.026}	0.63	23.81	23.56	51.94	0.026
GeAPO-5 _{0.031}	0.72	23.32	23.55	51.64	0.031
AlPO-18	0	23.25	24.47	52.28	0
low Si-AlPO-18	0.023	23.16	24.53	52.30	0.001

^a: measured by XRF.

Table S3. EXAFS fitting data.

Sample	CN (Ge-O)	R	σ^2	E_0	R-factor
		Å	10^{-3}Å^2	keV	%
GeAPO-18 _{0.025}	4*	1.77±0.02	4.3±1.9	5.33±2.9	0.024
GeAPO-18 _{0.025} -30h	4*	1.75±0.01	5.2±1.6	3.50±2.5	0.017

Ge-O path, $S_0^2 = 0.77$, CN = 4* (fixed), k-range from 3 to 12.2, R-range from 1 to 2.2.

Table S4. Acidity of GeAPO-18, SAPO-18 and MgAPO-18. ^a

Sample	Weak acid density	Medium strength acid density
	μmol/g	μmol/g
GeAPO-18 _{0.012}	161	35
GeAPO-18 _{0.017}	103	35
GeAPO-18 _{0.020}	157	41
GeAPO-18 _{0.025}	129	44
GeAPO-18 _{0.027}	113	53
SAPO-18 _{0.011}	179	26
SAPO-18 _{0.016}	281	52
SAPO-18 _{0.029}	331	73
MgAPO-18 _{0.019}	392	64
MgAPO-18 _{0.025}	437	82
MgAPO-18 _{0.029}	400	103

^a: Estimated according to NH₃-TPD in fig. S11, with the desorption peak temperature < 200 °C being attributed to the weak strength acid sites and the temperature > 200 °C to the medium strength acid sites.

Table S5. Comparison of syngas conversion performance between AlPO-18 and GeAPO-18 in combination with ZnCrO_x.^a

Zeotype samples	Si/Al or Ge/Al	CO conv., %	CO ₂ sel., %	CH ₄ in HCs, ^b %	C ₂ ⁼ -C ₄ ⁼ in HCs, %	C ₂ ^o -C ₄ ^o in HCs, %	C ₅ + in HCs %	Light olefins yield, %
AlPO-18	0	24	41	8	63	18	11	9
Low Si AlPO-18 _{0.001}	0.001	49	39	3	77	10	10	23
GeAPO-18 _{0.027}	0.027	85	32	2	83	9	7	48

^aReaction condition: ZnCrO_x/zeotype = 2 (mass), H₂/CO = 2.5, 430 °C, 6 MPa, 1500 mL/(g_{cat}·h).

^bHCs representing "hydrocarbons".

Table S6. Inductively coupled plasma optical emission spectrometry (ICP-OES) analysis of the Zn content of the fresh and used GeAPO-18.

	Zn content (ppm)
Fresh GeAPO-18 _{0.025}	4
Used GeAPO-18 _{0.025}	52

Table S7. Differential adsorption heat of NH₃ over GeAPO-18, SAPO-18 and MgAPO-18 measured by microcalorimetric method.

	GeAPO-18 _{0.025}	SAPO-18 _{0.016}	MgAPO-18 _{0.029}
Differential adsorption heat, kJ/mol	143	156	193

Table S8. NH₃ binding energy over different T-sites of GeAPO-18, SAPO-18 and MgAPO-18.^a

SAPO-18		GeAPO-18		MgAPO-18	
T-site	Energy, eV	T-site	Energy, eV	T-site	Energy, eV
T1-O5	-1.39	T1-O5	-1.26	T1-O5	-1.73
T2-O3	-1.39	T2-O3	-1.25	T2-O3	-1.72
T3-O7	-1.43	T3-O7	-1.22	T3-O8	-1.69
Average	-1.40	Average	-1.24	Average	-1.71

^a: see figs. S8 and S19 for the structures.

Table S9. C₂H₄ binding energy over different T-sites of GeAPO-18, SAPO-18 and MgAPO-18.^a

SAPO-18		GeAPO-18		MgAPO-18	
T-site	Energy, eV	T-site	Energy, eV	T-site	Energy, eV
T1-O4	-0.60	T1-O4	-0.58	T1-O5	-0.64
T1-O5	-0.60	T1-O5	-0.53	T1-O6	-0.70
T1-O10	-0.60	T1-O10	-0.58	T1-O9	-0.66
T1-O12	-0.60	T1-O12	-0.55	T1-11	-0.61
T2-O2	-0.71	T2-O2	-0.62	T2-O3	-0.65
T2-O3	-0.56	T2-O3	-0.49	T2-O4	-0.70
T2-O8	-0.76	T2-O8	-0.62	T2-O7	-0.76
T2-O11	-0.60	T2-O11	-0.54	T2-10	-0.68
T3-O1	-0.54	T3-O1	-0.51	T3-O1	-0.59
T3-O6	-0.57	T3-O6	-0.61	T3-O2	-0.71
T3-O7	-0.64	T3-O7	-0.59	T3-O8	-0.68
T3-O9	-0.58	T3-O9	-0.52	T3-12	-0.61

^a: See figs. S8 and S20 for the structures.

Table S10. Syngas conversion performance over ZnCrO_x-GeAPO-18_{0.025re} under different reaction conditions.^a

Temperature °C	GHSV mL/(g·h)	Pressure MPa	CO conversion %	CO ₂ selectivity %	C ₂ =-C ₄ = in HCs %	C ₂ =-C ₃ = in HCs %
420	1500	6	81	36	80	55
420	1500	8	82	35	80	54
420	3000	8	73	36	82	57
430	1500	4	75	37	83	60
430	4500	4	56	40	89	68
430	1500	6	84	32	83	56
430	4500	6	71	36	84	59
430	3000	8	82	33	81	54
430	4500	8	77	35	84	57
440	4000	8	80	33	80	54

^a: Reaction condition: ZnCrO_x/zeotype = 2 (mass), H₂/CO = 2.5.

Note that GeAPO-18_{0.025re} is a sample of reproducing batch.

Table S11. Syngas conversion performance over ZnCrO_x-SAPO-18_{0.016} under different reaction conditions.^a

Temperature °C	GHSV mL/(g·h)	Pressure MPa	CO conversion %	CO ₂ selectivity %	C ₂ =-C ₄ = in HCs %	C ₂ =-C ₃ = in HCs %
420	1500	6	83	35	72	42
420	1500	8	88	32	60	29
420	3000	8	81	33	68	39
430	1500	4	70	38	65	39
430	4500	4	45	41	77	54
430	1500	6	82	32	74	44
430	4500	6	70	37	75	46
430	3000	8	84	32	72	42
430	4500	8	78	34	69	39
440	4000	8	81	33	72	43

^a: Reaction condition: ZnCrO_x/zeotype = 2 (mass), H₂/CO = 2.5.

Table S12. Comparison of syngas conversion performance between GeAPO-11, GeAPO-5 and GeAPO-18 in combination with ZnCrO_x.^a

	Ge/Al ratio ^b	CO conv. %	CO ₂ sel. %	CH ₄ in HCs %	C ₂ ⁼ -C ₄ ⁼ in HCs %	C ₂ ^o -C ₄ ^o in HCs %	C ₅ ⁺ in HCs %
GeAPO-11 _{0.026}	0.026	19	44	37	8	24	30
GeAPO-5 _{0.031}	0.031	75	38	4	30	36	30
GeAPO-18 _{0.027}	0.027	85	32	2	83	9	7

^a: Reaction condition: ZnCrO_x/zeotype = 2 (mass), H₂/CO = 2.5, 430 °C, 6 MPa, 1500 mL/(g_{cat}·h).

^b: Measured by XRF. The results show that the type of AlPO molecular sieves is an important factor and the AEI topology with 8-MR pores is selective for synthesis of light olefins from syngas.